A Comparative Study of RAG Techniques for Image Captioning

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Part I

Data Exploration and Pre-processing

1.1 Introduction

Automatic image captioning is a computer vision task that aims to generate natural-language descriptions of images. This project studies how different Retrieval-Augmented Generation (RAG) techniques influence the quality, relevance, and visual grounding of generated image captions. RAG combines information retrieval with text generation. Instead of relying only on the knowledge learned during training, it retrieves relevant information from an external knowledge base to support caption generation

1.2 Problem Statement

The project investigates how RAG strategies can support image captioning and how retrieval quality influences generated captions. The assigned group studies the Self-RAG technique, within a shared comparative-study protocol.

1.3 Objectives of Phase 1

The main objective of this phase is to explore, process, and prepare the MS COCO 2017 dataset for the implementation of the RAG techniques.

This phase covers several activities:

- Exploring the MS COCO 2017 dataset and understanding its structure.
- Analyzing object categories, scene complexity, and caption characteristics.
- Verifying the integrity of images and annotations.
- Preprocessing images and captions.
- Establishing reproducible knowledge-base, validation, and test splits.
- Organizing the dataset and its associated metadata.
- Generating image embeddings and preparing the vector index for retrieval experiments.

Overall, this phase establishes a structured and reliable data foundation for the implementation and comparison of different RAG techniques.

1.4 Overview of the MS COCO 2017 Dataset

1.4.1 Dataset Description

The project uses the Microsoft Common Objects in Context (MS COCO) 2017 dataset, which contains real-world images showing different objects, scenes, and interactions.

The dataset provides image annotations, object categories, and natural-language captions. In this project, two types of annotations are used:

Caption annotations: These provide natural-language descriptions of the images.

Instance annotations: These provide information about objects in the images, such as their categories and bounding boxes.

1.4.2 Dataset Sources and Files

The project uses the following annotation files:

- captionstrain2017.json: Contains reference captions for training images.

- instancestrain2017.json: Contains object-instance annotations for training images.

- captionsval2017.json: Contains reference captions for validation images used as the test set in this project.

- instancesval2017.json: Contains object-instance annotations for validation images, when needed for the analysis.

The image files are downloaded separately from the annotation files. Each image is linked to its annotations using its unique image ID.

1.5 Statistical Exploration of the Dataset

1.5.1 Object Category Analysis

Object Category Distribution

This analysis examines the frequency of object categories in the MS COCO Knowledge Base. Each object annotation is counted according to its category ID using Python's Counter function. The category IDs are then mapped to their corresponding names using the COCO API.

The results are sorted in descending order to identify the most frequent categories. The following bar chart presents the 50 categories with the highest annotation counts.

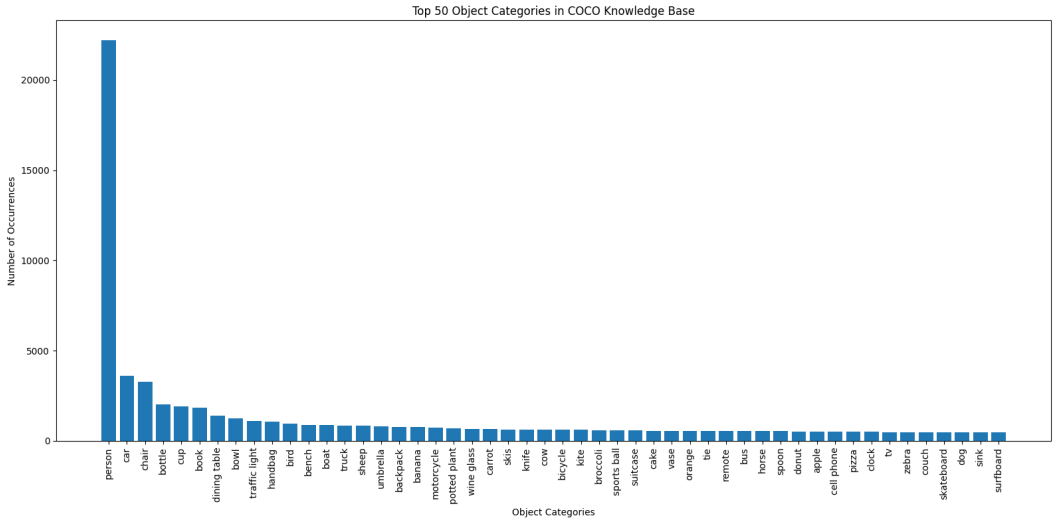


Figure 1.1: Top 50 Object Categories in the COCO Knowledge Base

Number of Images Containing Each Category

This analysis measures how many distinct images contain each object category in the Knowledge Base.

For each category, the corresponding image IDs are collected using a set to avoid counting the same image more than once. The number of unique image IDs is then calculated and added to the category statistics table.

This analysis provides additional information about the distribution of object categories across images, rather than simply counting the total number of object annotations.

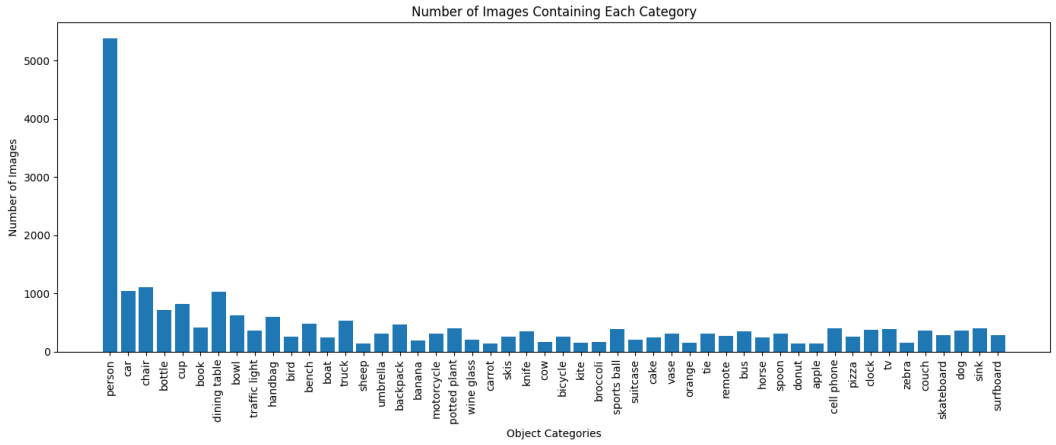


Figure 1.2: Number of Images Containing Each Category

1.5.2 Scene Complexity Analysis

Number of Objects per Image

This exploration aims to examine the number of objects present in each image of the MS COCO Knowledge Base. It provides an overview of how many objects typically appear in an image and helps identify differences in scene complexity.

Understanding the number of objects per image is important for this project because images with different numbers of objects may present different levels of difficulty for image captioning.

Distribution of Objects per Image

Scene Classification

1.5.3 Object Co-occurrence Analysis

Object Co-occurrence

Most Frequent Object Pairs

1.5.4 Caption Statistical Analysis

Caption Length Analysis

Caption Length Distribution

Vocabulary Size

Most Frequent Words

1.5.5 Image Color Analysis

Dominant Color Extraction

1.5.6 Dataset Statistics and Observations

1.6 Data Cleaning and Validation

1.6.1 Image Integrity Verification

1.6.2 Missing and Corrupted Files

1.6.3 Caption and Annotation Validation

1.6.4 Data Quality Summary

1.7 Data Preprocessing and Splitting

1.7.1 Image Preprocessing

1.7.2 Caption Preprocessing

1.7.3 Dataset Splitting

1.7.4 Data Organization and Metadata Preparation

1.7.5 Data Leakage Verification

1.8 Embedding Generation and Vector Indexing

1.8.1 CLIP Model and Configuration